

Paper Pulse

Weekly Research Paper Digest

Report Date: 2026-08-23 01:43 UTC

Papers Included: 20

Source: arXiv & IACR ePrint

Powered by DeepSeek AI

This digest presents the 20 most recent research papers automatically fetched, filtered, and summarized. Each entry includes the full title, authors, abstract, and AI-generated bilingual summaries.

Online: <https://090516mc.github.io/paper-pulse>

Table of Contents

1. Enforcing Winner-Only Disclosure: Verifiable Tally Hiding for Weighted DAO Governance
2. Multi-PGBF: Efficient Oblivious Key-Value Store and Application to Private Set Intersection
3. New Lower Bounds for Rows of $\mathbb{F}_2$ -Disjunct Matrices via Recursive Potentials
4. How Many Traces Suffice? PAC Guarantees for Profiled and Non-profiled Side-Channel Analysis
5. Rank Measures and Exponential Lower Bounds for Multilinear Secret Sharing
6. Constant-round MPC protocols with Fall-back Security
7. Circle-Linear Cryptanalysis: Bibrace Characters and Weak-Key Linear Distinguishers for CRAFT
8. Eavesdropper-Blind Remote State Preparation and Applications to Quantum Public-Key Encryption
9. Exact linear correlations and the cost of Walsh-transform key recovery, with application to SPEEDY
10. Two Novel Multidimensional Affine Variations of the Hill Cipher
11. Multidimensional Hill Cipher Substitution– Permutation Network
12. Pilaf: Fully Tight Two-Round Threshold Signatures with Adaptive Corruptions
13. Lightweight Lattice-based Single-Party Public-Key Authenticated Key Exchange
14. Midpoint Reset: A Full-Round Poseidon Collision from an Adaptively Chosen MDS Matrix
15. Revisiting the Transferability of Chosen- to Known-plaintext Attacks and Applications to Round-reduc...
16. $\mathcal{S}$: Prove-Phase Bounded-Memory Groth16 via Read-Write Streaming
17. Enabling Threshold Custody for the Lightning Network with Nested Threshold Multi-Signatures
18. Fully Homomorphic Encryption with Chosen-Ciphertext Security from LWE
19. G-CARL: Grounded Checklist-Aligned Reward Learning for Patient-Oriented Medical Report Interpretatio...
20. An Agentic Approach for Active Data Collection, Travel Behavior Modeling, and Weather-Sensitive Dema...

Paper 1

Enforcing Winner-Only Disclosure: Verifiable Tally Hiding for Weighted DAO Governance

Authors: Gongli Li

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1773>

Abstract

Token-weighted voting is widely used in DAO governance, but public voting weights together with weighted tallies can reveal identifiable voters' choices. Publishing only the final outcome reduces this disclosure, yet an output policy alone does not prevent a privileged participant from reconstructing the exact weighted tally during computation.

We present a verifiable winner-only tally-hiding construction for weighted binary voting. Registered weights are bound to credentials in zero-knowledge ballots, while weighted contributions remain encrypted through aggregation and comparison

against a public threshold. The blockchain adjudicates ballots, an off-chain backend performs the encrypted computation, and exact ciphertext and transcript bindings allow any public verifier to check that the published outcome corresponds to the accepted ballots. The only tally-derived plaintext output is the outcome bit.

The construction is parameterized by electorate size and contribution width; our prototype and formal transcript-privacy result deliberately study a bounded eight-voter, eight-bit instance with 134 encrypted gates and an actual three-of-five final release. For honest execution by all five trustees, we prove passive-public-observer backend transcript privacy from the accepted ciphertexts and outcome alone. Privacy against malicious sub-threshold trustees remains open.

Chinese Summary

研究背景与问题

在DAO治理中，代币加权投票被广泛采用，但公开的投票权重与加权计票结果可能泄露具体选民的投票选择。仅发布最终结果虽能减少披露，却无法阻止拥有特权的参与者在计算过程中重建精确的加权计票数据。

核心方法

本文提出一种**可验证的仅胜者公开（winner-only）计票隐藏方案**，适用于加权二元投票场景：

- **零知识凭证绑定**：注册权重与投票凭证绑定，生成零知识选票；
- **加密聚合与阈值比较**：加权贡献全程加密，通过聚合运算与公开阈值比较，仅输出最终结果比特；

-
链上链下协同：区块链负责裁决选票合法性，链下后端执行加密计算，并生成精确的密文与转录绑定，使任何公开验证者都能核验发布结果与已接受选票的一致性。

关键结果

-
方案参数由选民规模和贡献位宽决定，原型实现针对**8选民、8位权重**的受限实例，包含134个加密门电路，并实际完成5名受托人中3人参与的最终发布；

-
在全部5名受托人诚实执行的前提下，我们**形式化证明了被动公开观察者的后端转录隐私**——仅凭已接受密文和结果比特，无法推断任何额外计票信息。

局限与展望

针对恶意子阈值受托人的隐私保护问题仍待解决，这是未来研究的重要方向。

English Summary

Core Contribution

This paper presents a **verifiable winner-only tally-hiding construction** for weighted binary voting in DAO governance. Registered weights are bound to credentials via zero-knowledge ballots, while weighted contributions remain encrypted through aggregation and threshold comparison, ensuring the only plaintext output is the final outcome bit.

Method & Verification

The system combines blockchain-based ballot adjudication with an off-chain encrypted computation backend. Exact ciphertext and transcript bindings allow any public verifier to confirm that the published outcome corresponds to accepted ballots.

Key Results

For a bounded 8-voter, 8-bit instance (134 encrypted gates, 3-of-5 trustee release), we formally prove **passive-observer

backend transcript privacy** under honest execution by all five trustees. Privacy against malicious sub-threshold trustees remains an open challenge.

Paper 2

Multi-PGBF: Efficient Oblivious Key-Value Store and Application to Private Set Intersection

Authors: Siu-Ming Yiu

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1772>

Abstract

An oblivious key-value store is a data structure that can encode and decode n key-value pairs in a table of size m obliviously. After encoding, one cannot distinguish the encoded key-value pairs from other key-value pairs in the input domain. In this paper, we first propose a data structure called Peelable Garbled Bloom Filter (PGBF), which encodes the key-value pairs in a similar way to peeling and unpeeling an onion. Specifically, it can divide the key-value pair set (i.e., onion) as multiple subsets (i.e., peels) and order them from the outermost peel to the innermost peel by using a counting Bloom filter. However, using a small expansion rate (i.e., $\eta = m/n$) in PGBF will result in a non-empty core issue with non-negligible probability. To handle this issue, we propose Multi-PGBF by combining multiple PGBFs to do the peelings and unpeelings recursively. In addition, we propose a variant C-Multi-PGBF by clustering a large set into small sets to achieve faster encoding efficiency.

Our experiments show that Multi-PGBF and C-Multi-PGBF obtain the best encoding and decoding efficiency. Multi-PGBF improves the encoding time of RR (CCS' 22) by $65.1\% \sim 77.6\%$, while C-Multi-PGBF improves the encoding time of the clustered RR variant by $60.2\% \sim 64.7\%$. For decoding, Multi-PGBF is $28.6\% \sim 62.4\%$ faster than RR (CCS'22) and $89.7\% \sim 96.3\%$ faster than RB-OKVS (Usenix'23). When integrated into the state-of-the-art two-party and multi-party private set intersection protocols (Eurocrypt'21, Usenix'24), Multi-PGBF and C-Multi-PGBF lead to faster protocols than those using existing OKVS constructions in most settings.

Chinese Summary

背景与挑战：不经意键值存储（OKVS）是一种能够在不泄露信息的前提下，将 n 个键值对编码到大小为 m 的表中，且编码后无法区分有效键值对与输入域中的其他键值对。现有OKVS方案在编码效率或解码效率上存在瓶颈，难以同时满足高性能与低存储开销。

方法创新：本文首先提出**可剥落布谷鸟过滤器（PGBF）**，其编码过程类似“剥洋葱”——利用计数布隆过滤器将键值对集合（洋葱）划分为多个子集（洋葱皮），并从外层到内层依次剥离。然而，PGBF在较小扩展率（ $\eta = m/n$ ）下存在非空核心问题。为此，本文提出**Multi-PGBF**，通过组合多个PGBF递归执行剥离与反剥离，有效解决该问题。此外，为提升编码速度，进一步提出**C-Multi-PGBF**，通过将大集合聚类为多个小集合实现加速。

实验结果：实验表明，Multi-PGBF与C-Multi-PGBF在编码和解码效率上均达到最优。相比RR（CCS' 22），Multi-PGBF编码时间提升 $65.1\% \sim 77.6\%$ ，C-Multi-PGBF相对聚类RR变体提升 $60.2\% \sim 64.7\%$ 。解码方面，Multi-PGBF比RR（CCS' 22）快 $28.6\% \sim 62.4\%$ ，比RB-OKVS（Usenix' 23）快 $89.7\% \sim 96.3\%$ 。集成至最先进的双方及多方隐私集合求交协议（Eurocrypt' 21, Usenix' 24）后，在多数场景下均显著提升协议性能。

****意义****: 本文提出的OKVS结构为隐私计算领域提供了高效、可扩展的基础组件，尤其适用于大规模数据场景下的隐私集合求交等应用。

English Summary

****Core Contribution****: We propose Multi-PGBF, an efficient oblivious key-value store (OKVS) that recursively combines multiple Peelable Garbled Bloom Filters (PGBF) to eliminate the non-empty core issue at low expansion rates, plus a clustered variant C-Multi-PGBF for faster encoding.

****Key Results****: Experiments show Multi-PGBF improves encoding time by 65.1%–77.6% over RR (CCS’ 22), and C-Multi-PGBF by 60.2%–64.7% over the clustered RR variant. For decoding, Multi-PGBF is 28.6%–62.4% faster than RR and 89.7%–96.3% faster than RB-OKVS (Usenix’ 23). When integrated into state-of-the-art two-party and multi-party private set intersection protocols (Eurocrypt’ 21, Usenix’ 24), our constructions yield faster protocols in most settings.

****Impact****: This work provides a high-performance OKVS building block, advancing privacy-preserving applications requiring large-scale data processing.

Paper 3

New Lower Bounds for Rows of d -Disjunct Matrices via Recursive Potentials

Authors: Xiaopeng Zhao

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1771>

Abstract

In nonadaptive combinatorial group testing, given n items with at most d positives, the goal is to identify them using as few pooled tests as possible. A $t \times n$ binary matrix represents the design, where rows are tests and columns are items. The matrix is d -disjunct if no column is contained in the Boolean union of any d others. Let $T(d)$ be the minimum t for which such a matrix exists with $n > t$. Shangguan and Ge proved $T(d) \geq \frac{15 + \sqrt{33}}{24} d^2$ by counting private pairs (IEEE Trans. Inf. Theory, 62(12):7518-7521, 2016). In this paper, we strengthen their argument by introducing a column-deletion recurrence in which the light-heavy threshold varies with the recursive state $z = (n - t) / d^2$, rather than remaining fixed. This yields the improved bound $T(d) \geq 0.9283 d^2 - O(d)$. The analytic core reduces to a first-order ODE, and a self-contained interval-arithmetic certificate verifies that the solution reaches the required contact point.

Chinese Summary

研究背景与问题

在非自适应组合分组检测中，给定 n 个物品且其中至多有 d 个阳性，目标是用尽可能少的混合检测（池检测）来识别全部阳性。设计由 $t \times n$ 二元矩阵表示，行对应检测，列对应物品。若任意一列都不包含于任何 d 列的布尔并集中，则称该矩阵为 d -disjunct。记 $T(d)$ 为存在满足 $n > t$ 的此类矩阵所需的最小行数 t 。

已有结果与本文贡献

Shangguan 和 Ge 通过计数“私有对”证明了 $T(d) \geq \frac{15 + \sqrt{33}}{24} d^2 \approx 0.8643 d^2$ (IEEE Trans. Inf. Theory, 2016)。本文**强化**了这一论证，核心创新在于引入**列删除递推**，其中“轻-重”阈值随递归状态 $z = (n - t) / d^2$

动态变化，而非固定常数。这一改进使下界提升至：

$$T(d) \geq 0.9283d^2 - O(d)$$

技术方法

- **递归势函数**：通过列删除过程，将矩阵结构问题转化为一阶常微分方程（ODE）的初值问题。
- **解析核心**：递推的极限行为由 ODE 刻画，其解到达所需接触点（即达到最优下界）的路径被严格追踪。
- **数值验证**：采用**自包含的区间算术证书**（interval-arithmetic certificate）对 ODE 解的关键性质进行机器可验证的严格证明，避免依赖浮点近似。

主要创新点

1. **动态阈值**：首次在列删除递推中让阈值随递归状态变化，突破了固定阈值的瓶颈。
2. **严格验证**：区间算术证书提供了完全自动化的证明，无需人工分析误差。
3. **显著改进**：将常数从 0.8643 提升至 0.9283 ，缩小了与已知上界之间的差距。

意义

该结果推进了组合分组检测中下界理论的研究，为最优检测矩阵设计提供了更紧的约束，同时展示了递归势方法与计算机辅助证明结合的有效性。

English Summary

Core Contribution

This paper improves the lower bound for the minimum number of rows $T(d)$ in d -disjunct matrices used in nonadaptive combinatorial group testing.

Key Result

We prove $T(d) \geq 0.9283d^2 - O(d)$, significantly strengthening the previous best bound of $\frac{15+\sqrt{33}}{24}d^2 \approx 0.8643d^2$ by Shangguan and Ge (2016).

Method

Our approach introduces a **column-deletion recurrence** where the light-heavy threshold varies dynamically with the recursive state $z=(n-t)/d^2$, instead of remaining fixed. The analytic core reduces to a first-order ODE, and a self-contained **interval-arithmetic certificate** rigorously verifies that the solution reaches the required contact point, providing a machine-checkable proof.

Significance

This work demonstrates the power of combining recursive potential functions with computer-assisted verification, yielding a substantial constant-factor improvement and narrowing the gap to known upper bounds.

Paper 4

How Many Traces Suffice? PAC Guarantees for Profiled and Non-profiled Side-Channel Analysis

Authors: Fatemeh Ganji

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1770>

Abstract

Side-channel analysis (SCA) is commonly evaluated by reporting the number of traces required to reduce the rank of the correct key. Still, such evaluations remain empirical and do not explain how many traces suffice for reliable recovery, how profiling and attack data contribute separately, or when additional traces cannot overcome weak key distinguishability. We address these questions through a Probably Approximately Correct (PAC) formulation of profiled and non-profiled SCA. Our framework treats candidate-key scores as the common cryptanalytic object and separates finite-sample estimation from the intrinsic separation between the correct key and competing hypotheses. This distinction enables confidence guarantees for key rank and helps determine whether an attack failure is due to insufficient data or an inherently weak attack score. We instantiate the framework with representative profiled and non-profiled attacks chosen for their analytical tractability. Experiments on ASCAD-f and ASCAD-r show that this analytical tractability does not come at the cost of impractical attack performance. The profiled attack achieves exact recovery with tens of attack traces, whereas the non-profiled single-attack rank certificate guarantees exact recovery with about 1,000 traces. These results are competitive with recent ASCAD attacks and, in the non-profiled setting, substantially below the smallest trace counts identified in prior studies, while additionally providing finite-sample guarantees on key rank. More generally, the same finite-sample rank analysis can be adapted to other learners and distinguishers by deriving the corresponding score-gap guarantees. Overall, the framework turns trace complexity from an empirical attack observation into a reusable finite-sample criterion for key recovery.

Chinese Summary

研究背景与问题

侧信道分析（SCA）的评估通常依赖于报告恢复正确密钥所需的最小迹线数量。然而，这类评估本质上是经验性的，无法回答三个关键问题：究竟多少条迹线足以实现可靠恢复？建模数据与攻击数据各自贡献了多少？以及当攻击得分本身区分度不足时，增加迹线是否无效？

方法与框架

本研究提出一种基于**概率近似正确（PAC）学习理论**的统框架，用于建模侧信道攻击（包括有画像与无画像两类）。该框架的核心创新在于：

- 将候选密钥得分视为统一的密码分析对象，将**有限样本估计误差**与**正确密钥与竞争假设间的内在可分性**明确分离；
- 这一区分使得我们能够对密钥秩（key rank）提供置信保证，并诊断攻击失败的原因是数据不足还是得分本身区分度弱。

实验与结果

作者选取了具有解析可处理性的代表性攻击来实例化该框架，并在**ASCAD-f 与 ASCAD-r**数据集上验证：

- **有画像攻击**：仅需数十条攻击迹线即可实现精确恢复；
- **无画像攻击**：通过单次攻击的秩证书，约**1000 条迹线**即可保证精确恢复。

该性能与近期最优

ASCAD

攻击相当，且无画像场景下的迹线需求显著低于既往研究的最小值，同时额外提供了密钥秩的有限样本保证。

创新与意义

该框架将迹线复杂度从经验观察转化为**可复用的有限样本密钥恢复准则**，且可推广至其他学习器与区分器（只需推导相应的得分间隔保证）。这为侧信道分析提供了理论支撑，有助于更科学地评估攻击可靠性与数据需求。

English Summary

Core Contribution

This paper introduces a **Probably Approximately Correct (PAC)** framework for profiled and non-profiled side-channel analysis (SCA), transforming trace complexity from an empirical metric into a reusable finite-sample criterion for key recovery.

Key Innovation

The framework separates **finite-sample estimation error** from the **intrinsic separability** between the correct key and competing hypotheses. This distinction enables confidence guarantees on key rank and diagnoses whether attack failures stem from insufficient data or inherently weak score discrimination.

Main Results

Experiments on ASCAD-f and ASCAD-r demonstrate practical competitiveness: the profiled attack achieves exact recovery with tens of attack traces, while the non-profiled rank certificate guarantees recovery with $\sim 1,000$ traces—substantially below prior reported minima. The framework is also generalizable to other learners/distinguishers via corresponding score-gap guarantees.

Significance

This work provides the first PAC-style theoretical grounding for trace complexity in SCA, offering both confidence bounds and practical performance competitive with state-of-the-art attacks.

Paper 5

Rank Measures and Exponential Lower Bounds for Multilinear Secret Sharing

Authors: Shahram Khazaei

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1769>

Abstract

A multilinear secret-sharing scheme shares a vector secret and can therefore amortize share size over the secret dimension. This amortization can invalidate lower bounds proved for one-dimensional linear schemes, and the best previous explicit lower bound for multilinear schemes was quasipolynomial, $n^{\Omega(\log n)}$. We prove that the Razborov-Gal rank measure survives amortization: the normalized size of a multi-target monotone span program is at least the rank measure of the function it computes. Combined with the rank witnesses of Pitassi and Robere, this gives an explicit family of access structures for which every perfect multilinear scheme over every finite field has average and maximum information ratio $2^{\Omega(n)}$. The worst-case multilinear information ratio is therefore $2^{\Theta(n)}$, answering a question of Beimel. We further extend the bound to schemes whose sharing algorithm is arbitrary and whose reconstruction is affine-linear, under pairwise statistical privacy below one; combined with the degree-reduction theorem of Beimel, Othman, and Peter, this yields exponential normalized lower bounds for every

fixed reconstruction degree whenever the secret dimension is $2^{o(n)}$.

Chinese Summary

研究背景与核心问题

秘密共享方案中，**多线性方案**通过共享向量秘密，可在秘密维度上摊薄份额大小，但这使一维线性方案的下界失效。此前多线性方案的最佳显式下界仅为拟多项式 $n^{\Omega(\log n)}$ ，能否证明指数级下界是开放问题。

方法与主要结果

本文证明 **Razborov-Gál 秩度量在摊薄后依然成立**：多目标单调跨度程序的归一化规模不小于其计算函数的秩度量。结合 Pitassi-Robere 的秩见证，得到：

- 存在显式访问结构族，使任意有限域上所有完美多线性方案的平均及最大信息率均为 $2^{\Omega(n)}$ ；
- 因此最坏情形多线性信息率为 $2^{\Theta(n)}$ ，**解决 Beimel 提出的开放问题

扩展与推广

进一步将下界推广至：

- **共享算法任意、重构仿射线性**的方案，在秘密维度 $2^{o(n)}$ 且满足低于一的成对统计隐私条件下；
- 结合 Beimel-Othman-Peter 的降度定理，对**任意固定重构度**得到指数级归一化下界。

创新点

1. 首次证明秩度量对多线性摊薄的鲁棒性；
2. 将多线性方案下界从拟多项式提升至指数级；
3. 统一处理统计隐私与仿射重构，显著拓展适用场景。

English Summary

Core Contribution

This paper proves that the **Razborov-Gál rank measure survives amortization** in multilinear secret sharing: the normalized size of any multi-target monotone span program is at least the rank measure of the function it computes.

Key Results

- Combined with Pitassi-Robere's rank witnesses, we obtain an **explicit family of access structures** where every perfect multilinear scheme over any finite field has average and maximum information ratio $2^{\Omega(n)}$.
- This settles Beimel's open question, showing the worst-case multilinear information ratio is $2^{\Theta(n)}$ —an exponential improvement over the previous quasipolynomial bound.
- The bound extends to schemes with arbitrary sharing algorithms and affine-linear reconstruction, under pairwise statistical privacy below one, yielding exponential normalized lower bounds for every fixed reconstruction degree when the secret dimension is $2^{o(n)}$.

Significance

This is the **first exponential lower bound** for general multilinear secret sharing, demonstrating that amortization does not weaken rank-based arguments. The unified framework covers both perfect and statistical privacy, broadening the applicability of rank measures in cryptographic complexity.

Paper 6

Constant-round MPC protocols with Fall-back Security

Authors: Raghavendra Vernekar

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1768>

Source

Abstract

The notion of Best-of-Both-Worlds introduced in the work of Ishai et al. (CRYPTO 2006) investigated whether an MPC protocol can simultaneously provide two incomparable security guarantees depending on the number of corrupted parties. As a special case of this, Chaum et al. initiated the study of protocols that tolerate unbounded corruption within a certain adversary structure and PPT corruption of any number of parties beyond that. More recently, Acharya et al. (CRYPTO 2023) formalized this notion as MPC with fall-back security. Although the feasibility of such protocols has now been thoroughly studied in prior works, most of the existing protocols require round complexity linear in the number of parties and the computation size.

In this work, we study the round complexity of MPC with fall-back security in the threshold corruption setting, presenting constant-round protocols for optimal thresholds. We present a semi-honest fall-back secure protocol for $t < \frac{n}{2}$ with 3 rounds, in the plain model, whereas the best known protocol in the same setting takes at least 11 rounds. In the CRS model, we present a maliciously fall-back secure protocol for the same threshold with 4 rounds, satisfying unanimous abort (UA). Finally, we extend this to a 5-round protocol that satisfies fairness in the presence of unbounded adversaries for $t < \frac{n}{2}$ corruptions and UA tolerating PPT adversaries for arbitrary corruption beyond that. In the malicious setting, we construct the first constant-round fall-back secure protocols.

Chinese Summary

研究背景与问题

安全多方计算（MPC）协议通常针对固定数量的腐败方设定安全阈值。然而，现实场景中攻击者数量可能动态变化。为此，Ishai等人（CRYPTO 2006）提出“两全其美”概念，研究协议能否同时提供两种不可比较的安全保证。Chaum等人进一步探索了在特定 adversary 结构内容忍无界腐败、超出该结构则容忍PPT腐败的协议。最近，Acharya等人（CRYPTO 2023）将此形式化为**回退安全（fall-back security）**MPC。

现有挑战

尽管回退安全MPC的可行性已被充分研究，但现有协议的**轮复杂度**大多与参与方数量和计算规模呈线性关系。这严重制约了实际部署效率，尤其是在网络延迟较高的分布式环境中。

本文贡献

本研究聚焦于**阈值腐败设定**下回退安全MPC的轮复杂度优化，主要成果如下：

1. **半诚实模型**：在普通模型（plain model）中，针对 $t < n/2$ 的阈值，提出**3轮**回退安全协议。此前最佳协议需至少11轮，本文实现了近4倍轮数缩减。
2. **恶意模型**：在CRS模型中，针对相同阈值，提出**4轮**恶意回退安全协议，满足**一致中止（UA）安全性**。这是首个常数轮恶意回退安全协议。

3. **公平性扩展**: 进一步构建**5轮**协议, 在 $(t < n/2)$ 腐败下对无界敌手满足公平性, 同时对任意数量腐败的PPT敌手满足UA。

意义

本文首次证明回退安全MPC可在**常数轮**内实现, 突破了以往线性轮数的瓶颈。这一结果不仅理论价值显著, 也为实际部署提供了更高效的解决方案。所有协议均达到最优阈值, 且在半诚实与恶意设定下均实现了常数轮, 填补了该领域的关键空白。

English Summary

Core Contributions

This paper investigates the **round complexity** of MPC with fall-back security under threshold corruption, achieving **constant-round protocols** for optimal thresholds.

Key Results

- Semi-honest setting**: A **3-round** fall-back secure protocol for $(t < n/2)$ in the plain model, improving the prior best of 11 rounds by nearly $4\times$.
- Malicious setting**: A **4-round** protocol in the CRS model satisfying unanimous abort (UA) — the **first constant-round** maliciously fall-back secure protocol.
- Fairness extension**: A **5-round** protocol achieving fairness against unbounded adversaries for $(t < n/2)$ corruptions, while maintaining UA against PPT adversaries for any larger corruption set.

Significance

This work breaks the previous linear-round barrier, demonstrating that fall-back security is achievable in constant rounds. All protocols attain optimal corruption thresholds, providing both theoretical advances and practical efficiency improvements for secure computation in dynamic adversarial environments.

Paper 7

Circle-Linear Cryptanalysis: Bibrace Characters and Weak-Key Linear Distinguishers for CRAFT

Authors: Roberto Civino

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1767>

Source

Abstract

Linear cryptanalysis measures the correlation of a cipher with the characters of the group used to define differences. If that group is replaced by a second elementary abelian group structure on the same set, here the one coming from a binary bibrace, then the admissible masks are no longer the ordinary scalar products: exactly half of them survive, and the other half are forced to be quadratic. Beyne's geometric approach develops linear cryptanalysis over an arbitrary finite abelian group, providing a natural framework for this setting. We instantiate it on the group of a particular bibrace and apply it to Craft.

Over this group the Midori/Craft S-box has four probability-one relations, forming a small subgroup of the dual which the S-box preserves in both directions. Inside that subgroup a mask propagates deterministically and linearly, so the search for the best

trail is a minimum weight codeword problem, which we solve exactly by complete enumeration rather than heuristically.

A trail costs correlation, and it restricts the key to a weak-key class. The two are usually derived from the same data. We show that the correct reading, obtained by analysing the diffusion layer and the key addition together rather than separately, gives a class several bits larger than the one obtained cell by cell. One concrete consequence is that Craft's round constants, whatever their values, impose no restriction at all.

On Craft we obtain weak-key distinguishers up to eighteen rounds. At fourteen rounds the squared correlation is 2^{-44} over a class of 2^{108} keys, against $2^{-62.12}$ for the designers' linear hull, which is the best known linear result on the cipher and holds for all keys. On that class we therefore improve the best linear correlation by eighteen bits at equal round count, and we reach four rounds further than the best known linear hull. Both the distinguishers and the weak-key criterion are verified experimentally, with a negative control on random keys.

Chinese Summary

摘要

本文提出一种新型的**圆-线性密码分析**框架，将线性密码分析从传统有限域推广到由**二元双括号（bibrace）**结构定义的任意有限阿贝尔群上。在该群中，可用的线性掩码仅有传统标量积的一半，其余掩码被迫呈现二次形式。作者利用Beyne的几何方法，在特定的bibrace群上实例化该框架，并应用于轻量级分组密码**CRAFT**。

研究发现，Midori/CRAFT的S盒在该群上具有**四条概率为1的线性关系**，构成一个小子群，且S盒在两个方向上均保持该子群不变。在此子群内，掩码以确定性方式线性传播，因此寻找最优线性轨迹可转化为**最小重量码字问题**，并通过完全枚举而非启发式算法精确求解。

关键创新在于**密钥加与扩散层的联合分析**：传统方法将两者分开处理，而本文表明，正确的分析可得到比逐单元分析大数比特的弱密钥类。具体而言，CRAFT的轮常数无论取值如何，均不引入任何额外限制。基于此，作者在CRAFT上构造了**最高18轮的弱密钥区分器**。在14轮时，平方相关性为 2^{-44} ，对应 2^1 个密钥的弱密钥类，而设计者的线性壳在相同轮数下相关性为 $2^{-62.12}$ （对所有密钥成立）。因此，在该弱密钥类上，本文在相同轮数下将最佳线性相关性提升了**18比特**，并比已知最佳线性壳多推进**4轮**。所有区分器与弱密钥判断均通过实验验证，并以随机密钥作为阴性对照。

English Summary

Abstract

This paper introduces **circle-linear cryptanalysis**, a novel framework that extends linear cryptanalysis to arbitrary finite abelian groups defined by a **binary bibrace** structure. Only half of the traditional linear masks remain valid; the rest become quadratic. Instantiating the framework on a specific bibrace group, the authors analyze the lightweight cipher **CRAFT**.

The Midori/CRAFT S-box exhibits four probability-one linear relations forming a small subgroup preserved in both directions. Within this subgroup, mask propagation is deterministic and linear, reducing the best-trail search to a minimum-weight codeword problem solved exactly via complete enumeration.

A key innovation is the **joint analysis of key addition and diffusion**, yielding weak-key classes several bits larger than naive cell-by-cell analysis. Notably, CRAFT's round constants impose no additional restrictions. The authors construct weak-key distinguishers up to **18 rounds**; at 14 rounds, the squared correlation is 2^{-44} over 2^1 keys, versus $2^{-62.12}$ for the designers' linear hull (valid for all keys). This improves the best linear correlation by 18 bits at equal rounds and extends four rounds further. All results are experimentally verified with random-key negative controls.

Paper 8

Eavesdropper-Blind Remote State Preparation and Applications to Quantum Public-Key Encryption

Authors: Alexandru Cojocaru

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1766>

Abstract

Remote state preparation (RSP) is a central primitive in quantum cryptography, enabling classical parties to remotely construct quantum states using only classical communication. As a result, RSP serves as a key building block in numerous protocols involving classical clients and quantum servers, allowing classical parties to leverage the advantages offered by powerful quantum computers. All known constructions of RSP rely on strong cryptographic assumptions, typically variants of trapdoor claw-free functions (TCFs).

In this work, we initiate the study of a weaker form of remote state preparation, which we call eavesdropper-blind remote state preparation (EB-RSP). Informally, EB-RSP requires blindness only against external observers who see the transcript of the honest protocol, rather than against the quantum server itself. Despite this relaxed adversarial model, the resulting notion remains sufficient for useful cryptographic applications. In particular, we show that two-message EB-RSP already suffices to construct quantum public-key encryption with classical public keys and quantum ciphertexts. We then construct two-message EB-RSP protocols from specific one-way group actions, yielding a first step toward RSP-type primitives based on assumptions that do not rely on trapdoors. Finally, we observe that existing RSP constructions are likely naturally adaptable to the two-message EB-RSP notion; we demonstrate this explicitly for a concrete TCF-based RSP construction.

Chinese Summary

研究概述

远程态制备（RSP）是量子密码学的核心原语，允许经典方仅通过经典通信远程构造量子态，是连接经典客户端与量子服务器的重要桥梁。然而，现有RSP方案均依赖强密码学假设（如陷门爪函数）。

核心贡献

****1.**

新概念提出^{**}：首次定义^{**}窃听者盲远程态制备（EB-RSP）^{**}——仅要求对外部窃听者隐藏量子态，而非对量子服务器本身。这种弱化模型虽放宽了安全性要求，但仍足以支撑实际应用。

****2. 应用突破^{**}**：证明^{**}两消息EB-RSP^{**}即可构造带经典公钥、量子密文的量子公钥加密方案，大幅降低了RSP的应用门槛。

****3. 假设创新^{**}**：基于^{**}单向群作用^{**}（无需陷门）构造了两消息EB-RSP协议，为无陷门假设下实现RSP类原语开辟了新路径。

****4. 兼容性验证^{**}**：指出现有TCF-based RSP方案可自然适配两消息EB-RSP模型，并以具体构造为例进行了显式验证。

意义与展望

本研究首次系统探索了弱化安全模型的RSP，在降低密码学假设强度的同时保持了功能可用性，为量子公钥加密等应用提供了更轻量级的构建模块，是迈向基于简单假设的RSP原语的重要第一步。

English Summary

Core Contributions

This paper introduces **eavesdropper-blind remote state preparation (EB-RSP)**, a relaxed RSP notion requiring blindness only against external observers, not the quantum server. Despite the weaker adversarial model, we prove that **two-message EB-RSP suffices for quantum public-key encryption** with classical keys and quantum ciphertexts. We construct two-message EB-RSP from **one-way group actions** (avoiding trapdoor assumptions), and show existing TCF-based RSP protocols can be naturally adapted to this notion. This work provides the first trapdoor-free RSP-type primitive and demonstrates that meaningful cryptographic applications remain achievable under significantly weaker security requirements.

Paper 9

Exact linear correlations and the cost of Walsh-transform key recovery, with application to SPEEDY

Authors: Bing Sun

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1765>

Source

Abstract

When two S-box layers of a round are separated by no key addition, the round correlation is a signed sum over all compatible intermediate masks, not a product of layer correlations, so the product rule can fail in either direction. Our central finding is that evaluating this intra-round sum exactly changes the assessment of the published linear cryptanalysis of SPEEDY, whose two S-box layers are separated only by ShiftColumns. We first develop the linear cryptanalysis of this setting: an exact one-round algorithm with a decidable exactness condition for the product rule, a dependency-graph decomposition, a covering-number bound strengthening linear-trail weight bounds, and a Walsh-support criterion in which the affine dimension of that support, limited by the endpoint key masks, fixes the key-recovery transform cost. Potentials use the independent-round-key model; complexities are in equivalent encryptions. Applied to SPEEDY, these tools revise published results: a reported five-round mask sequence has exact correlation $2^{-90.0962}$, not $2^{-93.0147}$; the new bound raises the unrestricted five-round weight bound from 53.7714 to 62.2616 bits; and the full-round attack on SPEEDY-7-192 reported at time $2^{158.06}$ needs at least $2^{199.97}$ encryptions in the pruning class considered. For SPEEDY-6-192 we give a six-round known-plaintext attack (data $2^{169.84}$, time $2^{170.20}$, memory 2^{156}) and show that the attack class defined here contains no attack with data and time both at most 2^{128} , its time being at least $2^{136.302}$. The same exact evaluation also revises a four-round differential-linear correlation.

Chinese Summary

研究背景与核心问题

当一轮密码中的两个S盒层之间没有密钥加操作时，轮相关度是所有兼容中间掩码的带符号和，而非各层相关度的简单乘积，因此乘积规则可能双向失效。这一现象在SPEEDY分组密码的线性分析中尤为关键——其两轮S盒层仅由ShiftColumns分隔。

主要方法与理论贡献

本文系统建立了该场景下的精确线性密码分析方法，核心贡献包括：

- **精确一轮算法**：提出带可判定精确性条件的乘积规则，可判断传统近似是否成立；
- **依赖图分解**：通过依赖图结构降低中间掩码求和的复杂度；
- **覆盖数界**：强化线性轨迹权重下界，给出更紧的安全界；
- **Walsh支撑准则**：发现Walsh变换支撑的仿射维度由端点密钥掩码决定，进而固定密钥恢复变换的计算成本。

分析采用独立轮密钥模型，复杂度以等效加密次数衡量。

关键应用结果（针对SPEEDY）

应用上述工具后，多项已发表结果被修正：

- 一个已报道的五轮掩码序列的精确相关度应为 $2^{-90.0962}$ ，而非 $2^{-93.0147}$ ；
- 无限制五轮权重下界从53.7714比特提升至62.2616比特；
- 对SPEEDY-7-192的全轮攻击，在考虑剪枝类中，原报道时间复杂度 $2^{158.06}$ 实际至少需要 $2^{199.97}$ 次加密；
- 对SPEEDY-6-192，给出六轮已知明文攻击（数据 $2^{169.84}$ ，时间 $2^{170.20}$ ，内存 2^{156} ），并证明本文定义的攻击类中不存在数据和时间均不超过 2^{128} 的攻击，其时间下界为 $2^{136.302}$ ；
- 精确评估同时修正了一个四轮差分-线性相关度。

创新点与意义

本研究首次揭示了无密钥加间隔下轮相关度精确计算对密码分析结论的重大影响，提供了理论完备的分析框架，并为SPEEDY的安全性评估提供了更严格的证据。

English Summary

Core Contribution

This paper establishes an exact linear cryptanalysis framework for cipher rounds where two S-box layers are separated by no key addition, showing that the round correlation is a signed sum over intermediate masks—not a product of layer correlations. We develop four key tools: an exact one-round algorithm with a decidable product-rule condition, dependency-graph decomposition, a covering-number bound that strengthens linear-trail weight limits, and a Walsh-support criterion linking affine dimension to key-recovery cost.

Key Results for SPEEDY

Applying this framework revises published cryptanalysis of SPEEDY: a reported five-round mask sequence has exact correlation $2^{-90.0962}$ (not $2^{-93.0147}$); the unrestricted five-round weight bound rises from 53.7714 to 62.2616 bits; the claimed full-round attack on SPEEDY-7-192 at $2^{158.06}$ actually requires at least $2^{199.97}$ encryptions. For SPEEDY-6-192, we provide a six-round attack (data $2^{169.84}$, time $2^{170.20}$, memory 2^{156}) and prove no attack in this class has both data and time below 2^{128} , with a time lower bound of $2^{136.302}$. A four-round differential-linear correlation is also corrected.

Significance

Our work demonstrates that exact intra-round correlation evaluation can fundamentally alter security assessments, providing both a rigorous theoretical framework and tighter bounds for SPEEDY.

Paper 10

Two Novel Multidimensional Affine Variations of the Hill Cipher

Authors: Porter Eldridge Coggins

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1764>

Source

Abstract

Two novel symmetric multidimensional affine nested variations of the Hill Cipher are presented. The Hill Cipher is a block polygraphic substitution encryption scheme based on a linear transformation of plaintext characters into ciphertext characters. In the time since Hill first published his encryption scheme, variations, modifications, and improvements of theoretical and practical importance have been published every year indicating that the Hill Cipher is an active area of cryptography research. The first variation presented in this paper incorporated invertible key matrices of orders 2, 4, and 8 such that the matrix values of the 2×2 matrix rotate positions with each block of characters in a similar manner to the rotating letter wheels of a German Enigma Encoder, then results of the 2×2 key matrices output are passed to 4×4 key matrices, and 8×8 key matrix, 4×4 key matrices, and rotative-value 2×2 key matrices. The second variation is configured with invertible key matrices of orders 4, 8, and 16 without rotation of matrix values in a similar manner to the first variation. In both variations, plaintext characters of each block are operated on by exclusive-or (XOR) vectors prior to multiplication with the matrices to create the affine ciphers. Strengths, weaknesses, and other considerations are provided in the discussion. Two proposals are also argued with rationale for a more robust character set for encryption and the increase in modulus that the character set allows, and the possible advantages and disadvantages of affine XOR vectors.

Chinese Summary

摘要

本文提出两种新型的希尔密码（Hill Cipher）多维仿射嵌套变体。希尔密码是一种基于明文与密文字符线性变换的分组多项替换加密方案。自Hill首次发表该方案以来，每年都有理论及实践重要性各异的变体、修改与改进问世，表明希尔密码仍是密码学研究的一个活跃领域。

****第一种变体****采用阶数为2、4、8的可逆密钥矩阵，其运作机制类似于德国恩尼格玛密码机的旋转字母轮： 2×2 矩阵中的数值随每个字符块的位置旋转，其输出结果随后依次送入 4×4 矩阵、 8×8 矩阵、再经 4×4 矩阵及旋转值 2×2 矩阵处理。****第二种变体****配置阶数为4、8、16的可逆密钥矩阵，但矩阵值不进行旋转，其余结构类似第一种变体。在两种变体中，每个分组的明文字符在进入矩阵乘法之前，先与异或（XOR）向量运算，从而构成仿射密码。

讨论部分分析了两种方案的强度、弱点及其他考量。此外，本文还基于理由提出两项建议：一是采用更健壮的字符集以增强加密能力，并利用该字符集带来的模数提升；二是探讨仿射XOR向量的潜在优势与劣势。

English Summary

Summary

This paper presents two novel symmetric multidimensional affine nested variants of the Hill Cipher, a block polygraphic substitution scheme based on linear transformation of plaintext into ciphertext. The first variant employs invertible key

matrices of orders 2, 4, and 8, where the 2×2 matrix values rotate with each character block (akin to the rotating wheels of a German Enigma machine), with outputs cascading through 4×4 and 8×8 matrices, then back through 4×4 and rotating 2×2 matrices. The second variant uses orders 4, 8, and 16 without rotation. Both variants apply XOR vectors to plaintext blocks before matrix multiplication to create affine ciphers. The discussion covers strengths, weaknesses, and proposes a more robust character set with increased modulus, alongside analysis of affine XOR vector trade-offs.

Paper 11

Multidimensional Hill Cipher Substitution– Permutation Network

Authors: Ill

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1763>

Abstract

MD-Hill-SPN is the first Hill-based construction to combine a multi-tier diffusion mix layer, a memory-hard KDF, and a simultaneous multi-metric empirical evaluation. Two independent runs of the full metric suite yield: (a) full plaintext avalanche from round 1 (mean 63.97–64.67 of 128 bits, ideal 64); (b) the differential-probability sampling floor of 2×10^{-5} reached at round 4 (50,000 of 50,000 output differences distinct, both sessions); (c) algebraic-degree lower-bound saturation at the maximum observable value from round 1; (d) linear-bias indistinguishable from random (combined exceedance 4.40%, below the 4.55% noise floor); and (e) branch numbers at the Singleton (MDS) bound for every tier ($B = 5$ for 4×4 , $B = 9$ for 8×8 , $B = 17$ for 16×16), computed exhaustively over weight-1 inputs. MD-Hill-SPN therefore moves beyond theoretical construction to a construction that passes a defined empirical evaluation suite: avalanche, differential sampling, linear-bias probing, algebraic-degree lower bounds, and MDS branch numbers under single-key, known-plaintext conditions with fixed parameters, an evaluation no prior Hill cipher variant has reported in full.

Chinese Summary

研究概述

****MD-Hill-SPN**** 是首个将多层扩散混合层、内存困难密钥派生函数（KDF）与多指标经验评估相结合的密码构造方案。该方案通过两次独立完整测试，系统验证了其密码学强度。

Hill

核心发现

- ****雪崩效应****: 从第1轮起即实现完全明文雪崩（128位中平均翻转63.97–64.67位，理想值为64）
- ****差分概率****: 第4轮达到采样下限 2×10^{-5} （两次测试中50,000个输出差分全部唯一）
- ****代数次数****: 第1轮即饱和于最大可观测值
- ****线性偏差****: 与随机分布不可区分（综合超阈值4.40%，低于4.55%噪声底限）
- ****分支数****: 所有层级均达到Singleton（MDS）界（ 4×4 为 $B=5$ ， 8×8 为 $B=9$ ， 16×16 为 $B=17$ ），基于权重1输入穷举计算

创新与意义

MD-Hill-SPN 超越了纯理论构造，成为首个通过完整经验评估套件的

Hill

变体——涵盖雪崩、差分采样、线性偏差探测、代数次数下界及MDS分支数验证，且均在单密钥、已知明文、固定参数条件下完成。此前没有任何 Hill 密码变体报告过如此全面的评估结果。

English Summary

Key Contribution

****MD-Hill-SPN**** is the first Hill-cipher construction integrating a multi-tier diffusion layer, a memory-hard KDF, and simultaneous multi-metric empirical validation. Two independent full-suite runs confirm: (a) full plaintext avalanche from round 1 (mean 63.97–64.67 of 128 bits vs. ideal 64); (b) differential-probability sampling floor of 2×10 reached at round 4 (50,000/50,000 distinct output differences); (c) algebraic-degree lower-bound saturation from round 1; (d) linear bias indistinguishable from random (4.40% exceedance, below 4.55% noise floor); and (e) Singleton-bound (MDS) branch numbers for all tiers ($B=5/9/17$ for $4 \times 4/8 \times 8/16 \times 16$), exhaustively computed over weight-1 inputs. This work moves Hill ciphers beyond theoretical design to an empirically validated framework under single-key, known-plaintext, fixed-parameter conditions—a comprehensive evaluation no prior Hill variant has reported.

Paper 12

Pilaf: Fully Tight Two-Round Threshold Signatures with Adaptive Corruptions

Authors: Quan Yuan

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1762>

Abstract

Threshold signatures are deployed in settings where an adversary may run many concurrent signing sessions and corrupt signers adaptively. Two-round schemes make this especially delicate. Their first-round messages are independent of the signed message and can be preprocessed offline, so a later corruption must reveal randomness that is consistent with commitments already published in prior transcripts. Existing adaptive constructions address this tension by adding rounds, relying on algebraic or knowledge assumptions, or paying non-tight losses from guessing the corruption pattern, the decisive session, or the final transcript.

We construct TPilaf , the first two-round threshold signature scheme that combines partially non-interactive signing with a fully tight proof against adaptive corruptions. The scheme is pairing-free and is built in prime-order groups from the MDDH assumption in the random-oracle model. Its first-round messages can be generated offline, and any threshold set of signers can aggregate their second-round shares into a single publicly verifiable signature.

The proof combines two ingredients. First, we introduce a linearly homomorphic dual-mode commitment with targetable opening. This lets the simulator open an already fixed commitment to the aggregate target imposed by a later

Fiat-Shamir challenge. Second, we use profile-wise zero-sum masking with posterior completion. Corruption openings and signing responses are therefore sampled from the exact conditional law while values already visible to the adversary remain cached. Together, these tools enable a delayed branch-decision argument. The reduction waits until the adversary's own queries determine the last touched coordinate, completes only latent state, and then binds the forged hidden branch. The final bound has no combinatorial loss in the number of users, threshold, sessions, or corruption patterns, and contains only the explicit bad-event and assumption terms appearing in the theorem.

Chinese Summary

****背景与挑战****：门限签名常部署于敌手可并发运行多个签名会话、并自适应腐化签名者的场景。两轮方案在此类场景下面临特殊困难：其首轮消息与待签消息无关，可离线预处理，因此后续腐化必须揭示与先前已发布承诺一致的随机数。现有自适应安全构造通过增加轮数、依赖代数或知识假设、或付出因猜测腐化模式/关键会话/最终副本而导致的非紧损失来应对这一张力。

****本文贡献****：我们构造了

TPilaf ——首个将部分非交互式签名与完全紧的自适应腐化安全证明相结合的两轮门限签名方案。该方案无需配对，基于素数阶群中的 MDDH 假设，在随机预言机模型下安全。其首轮消息可离线生成，任意门限规模的签名者集合可将各自的第二轮份额聚合为单个公开可验证的签名。

****核心证明技术****：证明结合两个关键组件。

1. ****线性同态双模承诺（目标可开）****：允许模拟器将已固定的承诺打开到由后续 Fiat-Shamir 挑战所强加的聚合目标值。
2. ****逐轮廓零和掩码（后验补全）****：腐化开启与签名响应均从精确的条件分布中采样，同时保持敌手已可见的值不变。

这两项工具共同实现 ****延迟分支判定**** 论证：归约等待敌手自身查询确定最后触及的坐标，仅补全潜在状态，然后绑定伪造的隐藏分支。最终安全界在用户数、门限、会话数或腐化模式上 ****无组合损失****，仅包含定理中显式的坏事件项与假设项。

English Summary

We present TPilaf , the first two-round threshold signature scheme achieving fully tight security against adaptive corruptions while supporting partially non-interactive signing. The scheme is pairing-free, built on prime-order groups under the MDDH assumption in the random-oracle model, with offline-preprocessable first-round messages and aggregatable second-round shares. Our proof introduces two novel tools: a linearly homomorphic dual-mode commitment with targetable opening (enabling the simulator to open fixed commitments to later Fiat-Shamir challenges), and profile-wise zero-sum masking with posterior completion (ensuring corruption openings and signing responses follow exact conditional laws). These enable a delayed branch-decision reduction that eliminates all combinatorial losses in user count, threshold, sessions, or corruption patterns, yielding a bound containing only explicit bad-event and assumption terms.

Paper 13

Lightweight Lattice-based Single-Party Public-Key Authenticated Key Exchange

Authors: Nyiang Melchisedech Mbeng

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1761>

Source

Abstract

Authenticated Key Exchange (AKE) is a cornerstone of secure communication, especially in resource-constrained IoT environments where lightweight and post-quantum security are paramount. While lattice-based cryptography offers promising solutions, existing post-quantum AKE protocols often prioritize strong security notions, such as the use of an IND-CCA encryption scheme, incurring overheads incompatible with IoT devices. This raises a critical question: Can one-way security (OW), a weaker but potentially more efficient notion, suffice for secure AKE in the post-quantum era? We address this challenge by revisiting the ALIKE framework (ISO/IEC 29192-4), which achieves OW-CCA-based AKE using deterministic RSA. However, RSA's quantum vulnerability and the lack of lattice-based OW-CCA schemes hinder its applicability today. Our work bridges this gap through three key contributions. First, we prove that the Hash-Before-Encrypt (HBE) paradigm generically transforms deterministic OW-CPA schemes into OW-CCA-secure ones. We additionally present the Fujisaki-Okamoto transform and its security proof construction, providing a reference for understanding the efficiency advantages of the proposed HBE-based approach. Second, we modify Bai et al.'s efficient lattice-based OW-CPA scheme to a deterministic variant and rigorously prove its security. Third, we generalize the SPAKE framework to support our OW-CCA construction, enabling post-quantum AKE with minimal assumptions, implement and benchmark the resulting protocol, demonstrating state-of-the-art efficiency for lightweight, quantum-resistant AKE. By relaxing security requirements from IND-CCA to OW-CCA while preserving adaptive security we achieve a practical balance between robustness and performance, paving the way for deployable solutions in constrained environments like IoT and connected vehicles.

Chinese Summary

研究背景与挑战

认证密钥交换（AKE）是安全通信的基石，在资源受限的物联网（IoT）环境中尤为重要，此类场景对轻量级和抗量子安全有极高要求。尽管基于格（lattice）的密码学提供了有前景的解决方案，但现有后量子AKE协议往往优先追求强安全概念（如IND-CCA加密方案），导致计算与通信开销过大，难以适配IoT设备。因此，一个关键问题浮现：**单向安全性（OW）**——一种更弱但可能更高效的安全概念——能否在后量子时代满足AKE的安全需求？

研究方法与核心贡献

本研究通过重新审视ALIKE框架（ISO/IEC

29192-4）来应对上述挑战，该框架利用确定性RSA实现了基于OW-CCA的AKE。然而，RSA的量子脆弱性以及缺乏基于格的OW-CCA方案，阻碍了其现代应用。我们通过三项关键贡献弥补了这一空白：

1. ****HBE范式证明****：我们证明了**哈希-加密（Hash-Before-Encrypt, HBE）**范式能够通用地将确定性OW-CPA方案转化为OW-CCA安全方案。同时，我们给出了Fujisaki-Okamoto变换及其安全证明构造，作为理解HBE方法效率优势的参照。
2. ****确定性格基方案****：我们将Bai等人提出的高效格基OW-CPA方案修改为确定性变体，并严格证明了其安全性。
- 3.

****SPAKE框架泛化****: 我们将SPAKE框架泛化以支持我们的OW-CCA构造, 在最小化假设的前提下实现后量子AKE, 并完成了协议实现与基准测试, 展示了轻量级、抗量子AKE的****最先进效率****。

主要发现与意义

通过将安全要求从IND-CCA放宽至OW-CCA, 同时保留自适应安全性, 我们在鲁棒性与性能之间实现了实用平衡。实验结果表明, 该协议在物联网和网联汽车等受限环境中具备实际部署潜力, 为后量子时代的轻量级安全通信铺平了道路。

English Summary

This paper addresses the challenge of designing lightweight, post-quantum Authenticated Key Exchange (AKE) for resource-constrained IoT environments. We revisit the ALIKE framework (ISO/IEC 29192-4) and bridge its gap in the quantum era through three contributions: (1) we prove that the Hash-Before-Encrypt (HBE) paradigm generically transforms deterministic OW-CPA schemes into OW-CCA-secure ones; (2) we modify Bai et al.'s efficient lattice-based OW-CPA scheme into a deterministic variant with rigorous security proof; (3) we generalize the SPAKE framework to support our construction, achieving minimal-assumption post-quantum AKE. Our implementation benchmarks demonstrate state-of-the-art efficiency, showing that relaxing security from IND-CCA to OW-CCA—while preserving adaptive security—strikes a practical balance between robustness and performance, making it viable for deployment in constrained environments like IoT and connected vehicles.

Paper 14

Midpoint Reset: A Full-Round Poseidon Collision from an Adaptively Chosen MDS Matrix

Authors: Sunghyeon Jo

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1760>

Source

Abstract

We give an explicit compression collision for all 28 rounds of the KoalaBear Poseidon instance with parameters $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$, in the setting where the round constants are fixed before the MDS linear layer is chosen. The main problem is to construct a single linear layer that simultaneously controls two executions through both the full and partial rounds. We do this by tracking their midpoint and half-difference. In each two-round block, one prescribed image of the linear layer cancels the midpoint against the next round constant, so the following odd cubic S-box receives opposite states and resets the midpoint to zero. Two additional images are reused throughout the permutation to return the half-difference to the same one-dimensional subspace. The resulting trajectory constraints determine a linear layer, while a scalar recurrence closes the final difference under feed-forward. For the KoalaBear instance we obtain a collision in all sixteen output coordinates with an MDS matrix satisfying the prescribed linear-layer checks. The scalar construction reduces to low-degree equations and admits a family of parameter choices, so the collision is not an isolated instance. The result exposes an adaptive correlation between fixed round constants and a subsequently chosen linear layer that matrix-only checks do not capture.

Chinese Summary

研究概要

本文针对 KoalaBear Poseidon 哈希实例 (参数为 $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$) 给出了一个完整的 28

轮压缩碰撞攻击。该攻击发生在**轮常数固定之后、MDS 线性层尚未选定**的自适应选择场景下。

核心方法

攻击的关键挑战在于构造一个**单一的线性层**，使其能同时控制全轮（full rounds）和部分轮（partial rounds）中的两条执行轨迹。作者通过追踪两条轨迹的**中点（midpoint）**和**半差（half-difference）**来实现这一目标：

- **中点重置机制**：在每个两轮区块中，线性层的一个预设映射将中点与下一轮常数抵消，使得后续的立方 S-box 接收到相反的输入状态，从而将中点重置为零。
- **半差子空间约束**：复用线性层的两个额外映射，将半差始终限制在同一维子空间内。
- **标量递推闭合**：通过标量递推关系处理前馈（feed-forward）下的最终差分闭合。

主要结果

- 成功构造了一个满足所有预设线性层检查的**MDS 矩阵**，并在全部 16 个输出坐标上实现了碰撞。
- 该构造将问题归结为**低次方程**，并允许一族参数选择，表明碰撞并非孤立实例。
- 攻击揭示了**固定轮常数与后续选择的线性层之间**存在一种自适应相关性，这种相关性是仅检查矩阵本身无法捕获的。

创新点

这是首个在自适应选择 MDS 矩阵场景下，对完整 KoalaBear Poseidon 实例（含全部 28 轮）实现碰撞的工作，为 Poseidon 类哈希的安全性分析提供了新的视角和攻击范式。

English Summary

Key Contribution

This paper presents the first explicit full-round (28 rounds) compression collision for the KoalaBear Poseidon instance with parameters $(t, \alpha, R_F, R_P) = (16, 3, 8, 20)$, under an adaptive-chosen-MDS-matrix setting where round constants are fixed before the linear layer.

Method & Result

The attack constructs a single linear layer that simultaneously controls two execution paths by tracking their midpoint and half-difference. In each two-round block, a prescribed image cancels the midpoint against the next round constant, resetting it to zero, while two reused images confine the half-difference to a one-dimensional subspace. A scalar recurrence closes the final difference under feed-forward.

The resulting MDS matrix satisfies all prescribed checks and yields a collision across all 16 output coordinates. The construction reduces to low-degree equations and admits a parameter family, proving the collision is not an isolated case.

Significance

This work exposes an adaptive correlation between fixed round constants and a subsequently chosen linear layer that matrix-only verification cannot detect, highlighting a new security gap in Poseidon-style hash designs.

Paper 15

Revisiting the Transferability of Chosen- to Known-plaintext Attacks and Applications to Round-reduced AES

Authors: Wenying Zhang

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1759>

Source

Abstract

Differential-based attacks represent the best known results for many block ciphers. Such attacks usually demand that the adversary choose plaintexts (CP) or ciphertexts (CC) in subspaces to satisfy differential trails. However, many widespread modes of operation or applications prohibit the adversary from directly choosing inputs for the majority of primitive calls. While Biham and Shamir already suggested a straightforward trade-off for standard differential attacks in their work on the DES, studies on advanced differential-based types, such as impossible-differential, rectangle, or mixture attacks, have surprisingly received little attention so far.

In this work, we study applications of differential-based attacks in the random known-plaintext model (RKP) of the above. For the AES as the probably most widespread block cipher, we derive the best existing distinguishers and attacks in the RKP model on all versions, improving earlier results by at least one round. Interestingly, we show that Demirci-Selcuk meet-in-the-middle attacks with differential enumeration, which require much related data, can also be effective in that setting without approaching the full codebook too closely. For several of our attacks, we showcase differences between the models as trails that lead to the best known attack complexities under chosen data are suboptimal in the RKP model, and can be replaced by better trails. While our results do not threaten the security of the full AES, and their complexities are too high to represent any threats, we hope to inspire cryptographers to also consider attacks in the RKP for future attacks.

Chinese Summary

研究背景与动机

基于差分分析的攻击是许多分组密码目前最优的攻击手段。这类攻击通常要求攻击者能够**选择明文（CP）或选择密文（CC）**，以构造满足差分路径的数据。然而，在实际应用中，广泛使用的加密模式或协议往往禁止攻击者直接控制底层分组密码的输入。尽管Biham和Shamir早在DES分析中就提出了标准差分攻击在已知明文模型下的简单折中方案，但对于更高级的差分类攻击（如不可能差分、矩形攻击、混合攻击）在已知明文场景下的研究却鲜有涉及。

主要贡献与方法

本文系统研究了**随机已知明文模型（RKP）**下差分类攻击的适用性，并以最广泛使用的分组密码AES为对象，取得了以下核心成果：

- **改进所有AES版本的攻击结果**：在RKP模型下，我们推导出了目前最优的区分器和攻击，比此前已知结果至少**多攻击一轮**。
- **发现新适用攻击类型**：令人惊讶的是，原本需要大量相关数据的Demirci-Selcuk中间相遇攻击（结合差分枚举）在RKP模型下依然有效，且无需逼近全码本。
- **模型间差异分析**：我们揭示了攻击路径在两种模型下的**最优性差异**——某些在选择明文模型下最优的差分路径，在RKP模型下并非最佳，可被更优路径替代。

结论与展望

虽然我们的攻击复杂度尚不足以威胁完整AES的安全性，但本文首次系统展示了RKP模型下高级差分攻击的潜力，并呼吁密码学界在未来设计攻击时，也考虑随机已知明文这一更贴近实际应用场景的模型。

English Summary

Core Contribution

This paper revisits differential-based cryptanalysis under the **random known-plaintext (RKP) model**, where attackers cannot choose inputs. Focusing on AES, we derive the **best-known distinguishers and attacks** for all versions, improving prior RKP results by **at least one round**. Notably, we show that Demirci-Selcuk meet-in-the-middle attacks with differential enumeration remain effective in this setting without approaching the full codebook.

Key Findings

We demonstrate that optimal differential trails under chosen-plaintext models are **not necessarily optimal** in the RKP model, and better trails can replace them. While our attacks do not threaten full AES security, this work is the **first systematic study** of advanced differential attacks in the RKP model, encouraging cryptographers to consider this practical attack scenario in future research.

Paper 16

Sluice : Prove-Phase Bounded-Memory Groth16 via Read-Write Streaming

Authors: Hyunok Oh

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1758>

Abstract

We present Sluice , a read-write streaming Groth16 prover that reduces prove-phase random-access working memory from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$ once the CRS, QAP, and witness are materialized as private streams. It preserves the standard Groth16 interface: a proof of 3 group elements, 3-pairing verification, and unchanged verifier contracts.

Our key technical contribution is $\text{Split-Butterfly-Merge}$ (SBM), an NTT algorithm in the read-write streaming model with $\mathcal{O}(\log N)$ memory, $\mathcal{O}(N \log N)$ total I/O, and $\mathcal{O}(\log N)$ sequential passes over external storage.

Combining SBM with streaming sparse R1CS evaluation and chunked Pippenger MSM yields a verifier-compatible Groth16 proving path that exchanges RAM for sequential storage I/O and wall-clock time. Our prototype uses a fixed-window MSM engineering point; the measurements validate memory reduction and proof compatibility, while the theorem states the asymptotically tuned MSM schedule.

We implement Sluice over BN-254. Direct prove-only runs produce valid 128-byte proofs through $N=2^{25}$. The same-size bounded-memory comparison is at $N=2^{23}$: Sluice succeeds under an 8GB Linux cgroup cap, whereas the standard prover is killed under 8GB and 12GB caps and succeeds only at 16GB. These results position Sluice as a storage-rich, RAM-limited proving option rather than a replacement for optimized in-memory provers.

Chinese Summary

研究背景与问题

Groth16 是目前最广泛使用的零知识证明系统，但其证明生成阶段需要 $\mathcal{O}(N)$ 级别的随机访问内存（RAM），其中 N 为电路规模。当 N 达到 2^{25} （约3300万）时，标准实现需要超过16GB内存，严重限制了在内存受限设备（如边缘节点、嵌入式系统）上的部署。

核心方法

本文提出 **Sluice**，一种读写流式（read-write streaming）Groth16 证明器，其核心创新是 **Split-Butterfly-Merge (SBM)** 算法：

- 在读写流式模型下实现 NTT（数论变换），仅需 $\mathcal{O}(\log N)$ 内存
- 总 I/O 复杂度为 $\mathcal{O}(N \log N)$ ，外部存储顺序扫描次数为 $\mathcal{O}(\log N)$
- 结合流式稀疏 R1CS 求值与分块 Pippenger MSM，构建完整证明路径

Sluice 保持标准 Groth16 接口不变：3 个群元素的证明、3 次配对验证，验证者合约无需任何修改。原型基于 BN-254 曲线实现，采用固定窗口 MSM 工程优化。

主要结果

- **功能验证**：在 $N=2^{25}$ 规模下成功生成有效的 128 字节证明
- **内存对比**（ $N=2^{23}$ 规模）：
 - Sluice 在 8GB Linux cgroup 限制下成功运行
 - 标准证明器在 8GB 和 12GB 限制下均被终止，仅在 16GB 下成功

创新意义

Sluice 将证明阶段的随机访问内存需求从 $\mathcal{O}(N)$ 降至 $\mathcal{O}(\log N)$ ，用顺序存储 I/O 和墙钟时间换取 RAM 节省。它定位为“存储丰富、内存受限”场景下的补充方案，而非替代优化的内存内证明器。理论分析给出了渐近最优的 MSM 调度方案，实验验证了内存缩减与证明兼容性。

English Summary

Core Contribution

We present **Sluice**, a read-write streaming Groth16 prover that reduces prove-phase random-access memory from $\mathcal{O}(N)$ to $\mathcal{O}(\log N)$ once CRS, QAP, and witness are materialized as private streams. The standard Groth16 interface is preserved: 3-element proofs, 3-pairing verification, and unchanged verifier contracts.

Key Technique

The main innovation is **Split-Butterfly-Merge (SBM)**, an NTT algorithm in the read-write streaming model achieving $\mathcal{O}(\log N)$ memory, $\mathcal{O}(N \log N)$ total I/O, and $\mathcal{O}(\log N)$ sequential passes over external storage. Combined with streaming sparse R1CS evaluation and chunked Pippenger MSM, this yields a verifier-compatible proving path that trades RAM for sequential I/O and wall-clock time.

Results

Implemented over BN-254, Sluice produces valid 128-byte proofs up to $N=2^{25}$. At $N=2^{23}$, Sluice succeeds under an 8GB Linux cgroup cap, while the standard prover is killed under 8GB and 12GB caps, succeeding only at 16GB. These results position Sluice as a storage-rich, RAM-limited proving option rather than a replacement for optimized in-memory provers.

Paper 17

Enabling Threshold Custody for the Lightning Network with Nested Threshold Multi-Signatures

Authors: Matias Furszyfer

Source

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1757>

Abstract

The Bitcoin Lightning Network secures hundreds of millions of dollars, yet channel endpoints rely on vulnerable single online keys.

Although threshold signatures are routinely used to protect on-chain Bitcoin, no practical deployment has been possible for Lightning channels.

This is because thresholdizing a Lightning party requires nesting a threshold signature scheme inside of an established two-party MuSig2 protocol without altering its nonce exchange or message flow.

In this work, we resolve this limitation by formalizing nested threshold multi-signatures, a new cryptographic primitive for thresholdizing one participant inside a multi-signature protocol.

As an instance of this primitive, we present Iceberg, the first construction for nested threshold MuSig2 signatures.

Iceberg enables one side of a Lightning channel to operate as a t -of- n threshold group while appearing to the counterparty as a standard MuSig2 participant.

As a result, threshold custody can be deployed unilaterally on today's Lightning Network without requiring any modifications to Bitcoin, the Lightning protocol, or channel counterparties.

We prove the security of Iceberg, integrate a prototype into a production Lightning node, and benchmark its performance.

Our measurements show that thresholdizing a Lightning channel incurs only modest overhead, since a threshold group tolerating one corrupted member sustains over 93% of the payment throughput of an unmodified endpoint.

Chinese Summary

研究背景与问题

比特币闪电网络承载着数亿美元价值的资产，但其通道两端依赖单一在线密钥，存在单点故障风险。尽管门限签名已广泛用于保护链上比特币，却始终无法实际部署于闪电通道。根本原因在于：将闪电网络参与方门限化，需要在既有的两方MuSig2协议内部嵌套一个门限签名方案，同时不改变其随机数交换流程或消息传递结构。

核心贡献

本研究通过形式化定义**嵌套门限多签名**（Nested Threshold Multi-Signatures）解决了上述难题——这是一种在多签名协议内部将某一参与方门限化的全新密码学原语。基于该原语，我们提出了*Iceberg*，这是首个嵌套门限MuSig2签名的构造方案。

Iceberg的关键特性在于：

- 允许闪电通道的一侧以 t -of- n 门限组形式运行
- 对通道对手方而言，该门限组仍表现为标准MuSig2参与方
- 因此，门限托管可**单方面**部署于现有闪电网络，无需修改比特币协议、闪电协议或对手方节点

安全性与性能评估

我们严格证明了Iceberg的安全性，将其原型集成到生产级闪电节点中，并完成了性能基准测试。实验结果表明：门限化闪电通道仅引入适度开销——在容忍单个腐败成员的门限组配置下，其支付吞吐量仍能达到未修改端点的**93%以上**。

意义与影响

该工作首次实现了闪电通道的实用门限托管，为通道资金安全提供了抵御单密钥泄露的防护能力，同时保持与现有网络基础设施的完全兼容性。

English Summary

Core Contribution

This paper introduces **Iceberg**, the first practical construction for nested threshold MuSig2 signatures, enabling unilateral threshold custody deployment on today's Lightning Network without modifying Bitcoin, the Lightning protocol, or channel counterparties. The key innovation is formalizing **nested threshold multi-signatures**—a new primitive that thresholdizes one participant inside an established two-party MuSig2 protocol while preserving its nonce exchange and message flow.

Results & Impact

We prove Iceberg's security, integrate a prototype into a production Lightning node, and benchmark its performance. Measurements show that thresholdized channels sustain over **93% payment throughput** of unmodified endpoints when tolerating one corrupted member. This work eliminates the single-key vulnerability of channel endpoints, making threshold custody practically deployable for the first time in Lightning Network—a system securing hundreds of millions of dollars—with only modest overhead.

Paper 18

Fully Homomorphic Encryption with Chosen-Ciphertext Security from LWE

Authors: Willy Susilo

Categories: Cryptography

URL: <https://eprint.iacr.org/2026/1756>

Source

Abstract

We construct (1-hop) fully homomorphic encryption (FHE) schemes with chosen-ciphertext (CCA) security from the learning with errors (LWE) assumption in the standard model. Security of our construction only relies on the circular-secure LWE, which matches the assumptions needed for FHE with the basic chosen-plaintext security. Besides, the scheme achieves a security notion that is strictly stronger than the CCA1 security. Prior FHE schemes with even just CCA1 security require either the random oracle model or non-falsifiable assumptions.

The construction follows the well-known Naor-Yung double encryption paradigm. However, unlike previous works [Boneh et al., ITCS 2012; Canetti et al., PKC 2017; Manulis and Nguyen, Eurocrypt 2024], which employ general zero-knowledge succinct non-interactive arguments of knowledge (ZK-SNARKs), we design a special succinct argument to prove the validity of FHE ciphertexts. The succinct argument is constructed from batch arguments for NP and a new primitive called predicate extractable commitment, which may be of independent interest.

Chinese Summary

摘要

本文基于**带错误学习（LWE）**假设，在标准模型下构建了首个具备**选择密文（CCA）安全性**的（单跳）全同态加密（FHE）方案。方案的安全性仅依赖**循环安全LWE**，这与基本选择明文安全（CPA）的FHE所需假设一致，无需额外强假设。同时，方案实现了**严格强于CCA1**的安全概念，优于以往所有仅达到CCA1安全的FHE方案——后者要么依赖随机预言机模型，要么依赖不可验证假设。

方法与创新

构建沿用经典的**Naor-Yung双重加密范式**，但突破了以往工作（如Boneh等、Canetti等、Manulis和Nguyen）依赖通用**零知识简洁非交互论证（ZK-SNARK）**的局限。作者设计了**专用简洁论证**来证明FHE密文的有效性，该论证由两部分组成：

- **NP的批量论证（Batch Arguments for NP）**
- **新原语：谓词可提取承诺（Predicate Extractable Commitment）**

这一设计避免了ZK-SNARK的强不可伪造性要求，仅需LWE假设即可实现，且新原语可能具有**独立研究价值**。

意义

该工作填补了FHE在标准模型下实现CCA安全的理论空白，将安全假设降低到与CPA-FHE同等水平，为同态加密在安全计算、隐私保护等场景中的实际应用提供了更坚实的理论基础。

English Summary

Abstract

This paper constructs the first (1-hop) fully homomorphic encryption (FHE) scheme with **chosen-ciphertext (CCA) security** under the **learning with errors (LWE)** assumption in the standard model. Security relies solely on **circular-secure LWE**², matching the assumption required for CPA-secure FHE. The scheme achieves a notion **strictly stronger than CCA1**¹, whereas all prior CCA1-secure FHE schemes required either the random oracle model or non-falsifiable assumptions.

The construction follows the **Naor-Yung double encryption paradigm**³, but replaces the general ZK-SNARKs used in previous works with a **specialized succinct argument** for proving FHE ciphertext validity. This argument is built from **batch arguments for NP**⁴ and a new primitive called **predicate extractable commitment**⁵, which may be of independent interest. This novel approach avoids strong assumptions while maintaining CCA security, representing a significant theoretical advance in standard-model FHE.

Paper 19

G-CARL: Grounded Checklist-Aligned Reward Learning for Patient-Oriented Medical Report Interpretation

Authors: Shiao Xie, Siyu Chen, Jianwei Lv, Bo Yuan, Yujin Wang et al.

Categories: cs.CL, cs.AI, cs.CV

URL: <https://arxiv.org/abs/2608.20331v1>

Keywords: reinforcement, learning

Abstract

Personalized interpretation of medical reports has emerged as an increasingly important need among patients. Addressing this need requires both evidence-grounded medical factuality and context-dependent patient communication, yet existing medical

vision-language tasks do not adequately capture these dual requirements. To bridge this gap, we introduce Patient-oriented Medical Report Interpretation (PMRI), a novel open-ended multimodal generation task that requires models to explain medical reports in accurate and accessible language based on a user's query and dialogue history. These two objectives differ fundamentally in their verifiability, yet remain tightly coupled, making them difficult to optimize jointly under conventional supervised fine-tuning and holistic reinforcement learning paradigms. To address this challenge, we propose G-CARL, a grounded, checklist-aligned reinforcement learning framework that combines multi-source retrieval for atomic claim verification with context-aware, instance-specific weighted checklists for response coverage, providing structured supervision for factuality, user-demand satisfaction, and expression quality without constraining response diversity. We further construct MMedReport, a real-world PMRI benchmark, along with a clinician-designed three-dimensional evaluation protocol. Extensive experiments demonstrate that G-CARL consistently outperforms existing post-training baselines in overall quality, claim-level precision, and checklist recall. Pairwise preference evaluation by clinicians further confirms that G-CARL produces interpretations that are more accurate and better aligned with patient needs.

Chinese Summary

研究背景与问题

患者对医疗报告个性化解读的需求日益增长，这要求模型同时具备**基于证据的医学事实性**与**贴合语境的医患沟通能力**。然而，现有医学视觉-语言任务未能充分覆盖这两项要求。

核心贡献

1. 新任务：PMRI

提出**面向患者的医疗报告解读（PMRI）**，一个开放式的多模态生成任务，要求模型依据用户查询与对话历史，用准确且通俗的语言解释医疗报告。两项目标在可验证性上本质不同却又紧密耦合，传统监督微调与整体强化学习难以联合优化。

2. 新框架：G-CARL

提出**G-CARL（基于事实核对的清单对齐强化学习）**框架，核心创新包括：

- **多源检索原子声明验证**：将生成内容拆解为可验证的原子声明，结合多源检索实现细粒度事实核查

-

上下文感知的实例级加权清单：动态生成个性化检查清单，为事实性、用户需求满足度与表达质量提供结构化监督，同时不限制回答多样性

3. 新基准：MMedReport

构建真实世界PMRI基准数据集，并设计**临床医生主导的三维评估协议**（整体质量、声明级精确率、清单召回率）。

主要结果

大量实验表明，G-CARL在整体质量、声明级精确率与清单召回率上**持续优于现有后训练基线**。临床医生的成对偏好评估进一步证实，G-CARL生成的解读更准确且更贴合患者实际需求。

English Summary

Overview

We introduce **PMRI (Patient-oriented Medical Report Interpretation)**, a novel open-ended multimodal generation task requiring models to explain medical reports accurately and accessibly based on user queries and dialogue history. To address the coupled yet distinct objectives of evidence-grounded factuality and patient-friendly communication, we propose **G-CARL**, a grounded checklist-aligned reinforcement learning framework. It combines multi-source retrieval for atomic claim verification with context-aware, instance-specific weighted checklists, providing structured supervision without constraining response diversity. We also build **MMedReport**, a real-world benchmark with a clinician-designed three-dimensional evaluation protocol. Extensive experiments show G-CARL consistently outperforms existing post-training

baselines in overall quality, claim-level precision, and checklist recall, with clinician pairwise preference confirming its superior accuracy and alignment with patient needs.

Paper 20

An Agentic Approach for Active Data Collection, Travel Behavior Modeling, and Weather-Sensitive Demand Prediction

Authors: Narges Ahmadi, Yubo Jiao, Jônatas Augusto Manzoli, Jiangbo Yu, Luis Miranda-Moreno

Source

Categories: cs.AI, cs.CL

URL: <https://arxiv.org/abs/2608.20320v1>

Keywords: model, language, machine, learning, large

Abstract

Travel behavior research increasingly combines digital data collection with predictive modeling, yet these stages are often developed and evaluated separately. This study proposes a three-agent workflow integrating conversational data collection, structured data processing, and behavioral prediction. A chatbot-administered, image-augmented stated-preference survey collected mode choices from student commuters across five predefined weather scenarios, yielding 454 respondent-scenario observations. Weather-related associations were analyzed using a multinomial logit model, while logistic regression and random forest provided machine-learning benchmarks. Nine locally deployed large language models (LLMs), ranging from 2 to 35 billion parameters, were evaluated across four zero-shot prompt-and-context conditions and extended through persona, few-shot, and vision-based configurations. Random forest achieved 69.6% five-class accuracy, while the best text-only zero-shot LLM reached 69.9% without task-specific fitting. Habitual travel information produced the most consistent gains, Expert framing generally outperformed Role-Play, and persona information was most useful when habitual travel information was unavailable. Few-shot prompting improved prediction for several models, with gains stabilizing after a small number of examples. Using the same weather images shown to respondents, the best vision-based configuration reached 71.5% five-class accuracy, indicating that visual context may provide additional predictive information for selected models. Overall, the study shows how conversational surveys, structured data processing, conventional behavioral modeling, machine learning, and multimodal LLM prediction can be coordinated within an auditable multi-agent workflow.

Chinese Summary

研究概述

本研究提出了一种**三智能体 workflow**，将对话式数据收集、结构化数据处理与行为预测整合于同一框架，解决了出行行为研究中数据采集与建模环节长期割裂的问题。

方法

-

数据收集：通过聊天机器人管理、图像增强的陈述性偏好调查，在五种预设天气场景下收集学生通勤者的出行方式选择，共获得454条受访者-场景观测数据。

- *建模对比*：采用多项Logit模型分析天气关联因素，并以逻辑回归和随机森林作为机器学习基准。

-

多模态LLM评估：部署9个本地大语言模型（参数规模2B–35B），在四种零样本提示条件下测试，并扩展了角色设定、少样本学习和视觉输入配置。

主要发现

- 随机森林五分类准确率为**69.6%**，最优纯文本零样本LLM达到**69.9%**（无需任务特定微调）。
- **习惯性出行信息**带来最稳定的性能提升；专家框架普遍优于角色扮演；当缺乏习惯性信息时，人物画像数据最为有效。
- 少样本提示对多个模型有增益，且收益在少量样本后趋于稳定。
- 使用受访者所见的相同天气图像，最优视觉配置达到**71.5%**的准确率，表明视觉上下文可为部分模型提供额外预测信息。

创新点

本研究首次系统性地将**对话式调查、结构化数据处理、传统行为模型、机器学习与多模态LLM预测**协调于一个可审计的多智能体流程中，为天气敏感的出行需求预测提供了新范式。

English Summary

Core Contribution

This study introduces a **three-agent workflow** that unifies conversational data collection, structured processing, and behavior prediction—addressing the traditional disconnect between survey design and modeling in travel behavior research.

Methods & Results

A chatbot-administered, image-augmented stated-preference survey captured mode choices from student commuters under five weather scenarios (454 observations). Nine local LLMs (2B–35B parameters) were benchmarked against multinomial logit, logistic regression, and random forest. The best text-only zero-shot LLM achieved **69.9%** five-class accuracy (matching random forest at 69.6%), while the top vision-based configuration reached **71.5%** using identical weather images shown to respondents. Habitual travel information yielded the most consistent gains; expert framing outperformed role-play; and few-shot prompting improved several models with rapid saturation.

Significance

This work demonstrates the first auditable, multi-agent coordination of conversational surveys, conventional econometric models, machine learning, and multimodal LLM prediction—offering a replicable framework for weather-sensitive demand forecasting and broader behavioral research.
